Course Prefix and Number: WLD-II
Instructor: Lisa -Moran
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Course Title: Welding Technology & Careers 2
Curriculum: Welding & Manufacturing Technology
Dual Credit: WLDG 110 & WLDG 120
Course Description:
WLD-II is intended for students seriously considering a career in the Welding & Related Industries. This is an intermediate course which enhances student skills & understanding of Industrial Safety,Shielded Metal Arc Welding (SMAW), Welding Procedure Specifications, Welder Qualification requirements, basic Welding Metallurgy, Precision Measurement, practical applications of basic Shop Drawing & Blueprints and Advanced Welding Symbols. Class time will be spent exploring professional standards, ongoing education options, requirements for Apprenticeships, career pathways, career readiness indicators, and professional networking strategies.
Students will receive hands-on training in basic & intermediate welding applications using the Shielded Metal Arc Welding (SMAW) process,as well as Oxy/Fuel Cutting and the use of power tools used in the preparation and repair of welds. Additional instruction will cover the procedures and equipment involved in the Destructive Testing of welds to Industry Specifications for Welder Qualification. During project based exercises students in WLD-II will operate under reduced direct supervision and will be expected to apply all previous information obtained in WLD-1, self-organize, set up equipment, and perform tasks in a manner typical of apprentices and entry-level welders.
Classwork will be split between classroom lecture, direct instruction and lab/shop exercises involving individual and group tasks.
Course Objectives:
A student completing this course will be capable of identifying and pursuing various positions, education opportunities, & career pathways in the Welding, Manufacturing and Skilled Trade sectors. Students are expected to develop a career plan and follow up with opportunities prior to graduation.
Upon course completion a graduate should be able to use the tools and equipment needed to competently prep & weld Groove & Fillet joint configurations in the flat, horizontal, vertical & overhead positions. Students will be able to measure, mark & cut mild steel using Oxy/Fuel or mechanical cutting systems, and prepare cut metal using finishing tools.
Graduates of this course will be capable of independently preparing, assembling and welding to codes & specifications, such as those defined by the AWS D1.1: Code for Structural Welding of Carbon Steel, and will be able to identify various metals & alloys common in welding & fabrication settings. Throughout the course the student will be working on lab welding projects, advancing progressively under instructor

evaluation, in preparation for the Mechanical Destructive Testing of their welds to prepare for Welder Performance Qualification on the job. Qualifications are performed using mild steel up to 3/8” thickness per AWS B4.0 guidelines.

Student materials:
A. Textbook (provided): G&W, Modern Welding-12th Edition: Bowditch, Turnquist & Althouse
B. Notebook: note taking is a compulsory portion of classroom education. Students are required to take notes during lectures and presentations.
C. Computers: Students will use laptops or Chromebooks provided by their home school. Laptops can be provided during class at Wilco if there are network or technical difficulties, or other extenuating circumstances. No personal computers are allowed on school networks.
Student Evaluation & Testing Policy
Students will be assessed through a combination of academic Testing, practical (hands-on) Skills Assessment and Participation in organized learning activities. Criteria for Skills Assessment are aligned to the American Welding Society guidance for Entry-Level Welders & AWS B1.11. Assignments are graded on a point value system aligned to the Wilco standard for grading.
Participation & Job Readiness evaluations are daily and ongoing assessments of student performance as it pertains to the Non-Technical skills & overall preparedness for job performance. Each student will be given a 400-point total each week (aligning with the standard “1-5” skills assessment model used in industry). Points will be reduced based on failure to participate, unsatisfactory performance or violations of policy & procedure.Student violations of Policies or Procedures are interpreted against Industry Best Practices, Guidelines and Codes. OSHA, AWS, ANSI and other Regulatory Organizations set the basis for student best practices and School Rules set the basis for student behavior. Removal by an instructor from class activity or unwillingness to participate will result in zero points earned for the day’s activity.
The Semester Exams & Finals will contain a written Exam as well as a Practical Test, making up 20% of the overall grade.
The remaining 80% of the Progress Scores are a combination of Hands-On/Technical Skill Development (30%), Participation/Preparedness/Job Readiness (40%), Practical: Quiz/Tests (10%)
Wilco Area Career Center Grading Scale:
100–90 = A
89 – 80 = B ■
79 – 70 = C
69 – 60 = D ■
■

below 59 = F

Dual Credit Evaluations:
Students will receive Dual Credit through Joliet Junior College if they have completed the minimum of required tasks by the end of their respective semester using the JJC grading standard (which may not directly correspond to the Wilco grading scale or timetable). If a student is not meeting the JJC requirement by the end of the respective Semester they may be removed from the JJC roster for Dual Credit.
JJC Grading Scale:
100-94 = A
88-93 = B
80-87 = C
79 & Below = DROP FROM DUAL CREDIT

Note: All in-class assignments must be actively worked on to their completion within the Welding Laboratory. For academic testing, re-tests are allowed based on student needs and instructor approval. Re-test dates shall be presented to students. Re-test on Practical/Lab based assessments may be subjected to reduced scores for late submission per instructor discretion.

Safety Policy & Procedure Overview:
Notice Arc Welding processes possess inherent and unavoidable health risks. Proper use of equipment and workspaces can eliminate or greatly reduce most risks. Regardless of compliance with safety guidelines, students will be exposed to the following risks: high heat, sparks, fire, high voltage electricity,
flying particles, moving or rotating powered equipment, sharp surfaces and falling objects. These risks can never be reduced to 0% and students must participate in shop activities for class credit.
Safety Guidelines and Personal Protective Equipment requirements are based on OSHA: cfr1910 & AWS/ANSI: Z49.1 standards. All safety policies are compulsory and compliance is mandatory for class credit. Students are required to abide by all shop safety criteria established by the instructor or be withdrawn from class activity. A detailed explanation of Shop Safety Policies & Procedures will be delivered during orientation. It is expected that students consistently abide by all safety directives while involved in class activities. Safety glasses will be worn at all times, except during sit-down lectures in the classroom area or during select times when the lab has been designated as safe.

After completion of the Safety Orientation training it is expected that students follow the guidelines established during training.Students are responsible for their own safety as well as the safety of other students, staff and visitors. Lab dress code requirements are a class safety policy and students improperly dressed for class will not be allowed access to lab areas.The unauthorized use of tools and equipment is strictly prohibited. Disruptive behavior, “Horseplay”, leaving work areas without authorization, or improper actions of any kind are considered a safety hazard and are grounds for disciplinary intervention.

Any injury or equipment damage shall be immediately reported to welding staff, regardless of severity. All safety violations are subject to review. Intentional or neglectful violations of safety procedures are subject to disciplinary intervention.

Heat Advisory Welding Shop temperatures can significantly exceed outdoor temps, which can lead to high temps in work areas. During hot days students are encouraged to bring a water bottle (or other form of hydration) to class. There are refill stations and water bottles are available for purchase at Wilco, but it
is recommended that students are properly hydrated upon arrival. Any indications of heat related stress should be immediately reported to welding staff

Dress Code:

Participation in Lab activities will require students to be properly dressed in clothing appropriate for an Industrial environment where exposure to heat, sparks, flames and falling objects are present. Students will be asked to wear clothing made from natural fibers such as cotton (or wool) and no synthetic materials such as polyester, spandex or vinyl. Recommendations are Cotton t-shirts, jeans and leather work-boots (steel or safety-composite toe boots are recommended) as daily wear items. Students will be expected to arrive “dressed for work” each day. Standards for professionalism apply to clothing as well, and students wearing items offensive or otherwise inappropriate for a work environment will be subject to disciplinary intervention and removal from class activity.

Personal Protective Equipment (PPE):

The use of Arc Welding equipment will require the student purchase of PPE. Students are required to provide a Welding Helmet, Welding Jacket & Welding Gloves. A basic pair of safety glasses will be issued, but students are required to replace missing or damaged safety glasses. Information about PPE specifications and requirements (as well as how & where to purchase) will be given during class orientation.

*Students on the Free/Reduced Lunch Programs will be issued a set of PPE by Wilco, to be returned upon course completion (See home school for details on FRL program)

Attendance:

Attendance and participation at Wilco is viewed as a Workplace Readiness Indicator and it is expected that students maintain a positive attendance standing. Students are evaluated on participation and attendance as part of their Performance Evaluations in order to ensure that all benchmark task objectives are not only completed, but completed within industry acknowledged timetables. Excessive absence in the workforce is a major reason for dismissal and student attendance habits may be verified by future employers reaching out to the Career Center for alumni references. Wilco will send a warning letter home to notify students and parents that five days of school have been missed. After the tenth absence, a parent conference may be required, along with a reduction in letter grade and possibly resulting in an Administrative Attendance Contract. Continued absences may result in the student being dropped from Wilco, and the student receiving a failing grade. The only absences not included in the policy are those

covered by a doctor's note, court document, or notification/verification of a family emergency or a death in the family.
· Excused absence: Students may make up work for credit but will have one day for every day absent from class plus one day to submit it.
· Unexcused absence: No credit for course work missed.
· Tardy to class: If a student is not in class by the late/second bell, the student will be considered tardy. The teacher will notify administration with a possible detention issued.
· School events: Students may be required to attend assemblies, field trips, and meetings at their home schools. Absence due to a school event is not factored into their Wilco absences.
· Suspensions: Students who are not in school because of disciplinary action may submit work in accordance with the excused absence policy.

Laboratory Maintenance:
Students are required to participate in laboratory maintenance (clean up) per shop policy/procedure. Maintaining shop organization and "housekeeping" is a routine part of the Industrial profession and Safe Shop Practices. If a student does not participate in laboratory maintenance during class periods points will be deducted from Class Participation scores. Clean up is inclusive of all workstations, shop floors and all equipment areas. Students are required to report any area or equipment found in an unsafe condition. Clean Up procedures will be detailed in class, posted in work areas and applied daily.

Student Code of Conduct:
Wilco Area Career Center is an educational extension of the home school. As an extension, Wilco maintains the policies of each school in combination with the policies developed by the Wilco Board of Control. The combined policies represent the discipline procedures that will be followed by the Wilco Staff, Faculty and Administration. The home school will be consulted/contacted on all discipline incidents. Disciplinary consequences will be the result of communication between the home school and Wilco's administration.
Three principles govern all discipline and regulations at Wilco Area Career Center:
1. Conduct that is disruptive to the educational environment is prohibited.
2. Conduct that infringes on the rights of others is prohibited.
3. Conduct that is unsafe is prohibited

Student Conduct expectations in the Welding Program are aligned to industry standards and students are expected to behave as if “they are on the job”. This means that basic workplace decorum and professional conduct is to be maintained while in class. Students are expected to be, at minimum, polite & courteous when interacting with staff, peers and visitors and respectful of the learning environment.
Students shall NOT:

●■
Violate procedures outlined in the Student Handbook
●■
Bring in weapons or fabricate dangerous objects from shop materials
●■
Leave class/work areas without approval
●■
Remove PPE in Shop areas
●■
Modify, damage, remove or improperly use materials, tools or equipment
●■
Exit/enter through unapproved doors, prop open doors or allow unauthorized persons access to the building.
●■
Violate school driving policies (authorized transportation to & from Wilco only)
●■
Carry flammables, valuables or electronic devices in pockets into the Shop Areas

*The Welding Program has a Zero Tolerance policy on fighting, wrestling and/or “Horseplay”. As an Industrial Worksite, safety hazards must be considered at all times, and any risky or violent behavior is extremely dangerous, and therefore STRICTLY prohibited at all times.

**District Schools strictly prohibit the use of Cell Phones in class. Wilco Students are not to use cell phones for any reason (unless clearly approved by School Administration). Use of cellular & bluetooth equipment (i.e. phones, speakers, and/or ear-buds) is not permitted in the lab areas due to the risk of signal interference that can damage welding equipment. Earbuds & Headphones are strictly prohibited at all times in hallways, classrooms and labs (unless clearly approved by School Administration).

This list is not exhaustive and additional Policies & Procedures will be detailed in class and posted in work areas. Students are responsible for becoming familiar with all Policies & Procedures, and they shall make an effort to abide by them. Students in violation of Program policy or procedure will receive corrective instruction. Repeat violations may result in ongoing Disciplinary intervention

Disciplinary Procedures:

Student compliance with Wilco and Welding Program rules is expected, and infractions will result in some form of Academic or Administrative Intervention. All efforts will be made to resolve issues directly with the student through conversation and in-class strategies, however, when there are recurring or significant issues that can not be resolved by conventional approach, Instructors may begin Intervention Procedures.
After a Verbal Redirection has been given and the behavior or condition persists, Instructors will:

1.■ Document the behavior, adjust Performance Assessment scores
2.■ Contact a parent to discuss behaviors and review strategies
3.■ Report issues to school counselors & administrators and develop a Corrective Action Plan.
4.■ Upon implementing Action Plan disciplinary actions can include Student Contracts, Suspension and/or Expulsion from the program (depending on nature & severity).

E-Learning:

Students will be issued an email address (@wilcoacc.org domain) for all E-Learning activities. All communication and assignments must be completed using the "@wilcoacc.org" Profile.
E-learning & Remote activities will be used when in-class instruction is not authorized or available. All E-learning assignments are required coursework and must be completed by assigned dates for full credit. Late submission scores will be reduced by 10%. Confirmed Sick-Days will be added to the deadline date.Instructors will provide instructions for live meetings during class time when conditions require remote learning

Dual Credit Alignment with Joliet Junior College (JJC):
Students have an opportunity to continue earning college credit through Joliet Junior College Welding Technology Program. WLD-2 covers JJC course requirements for the WLDG 110 (SMAW 1: Flat & Horizontal, 3 credits) and WLDG 120 (SMAW 2: Vertical & Overhead, 3 credits) Earning credits is dependent on the student meeting course requirements for WLD-1, applying for admission and completing the registration process, the student fulfilling college prerequisites, and the student meeting the academic rigor of the course.
· Dual Credit requirements are set by the participating college. Students must meet all placement requirements before signing up for the college course.
· Students may participate in the coursework at Wilco and decline the dual credit. However, the work assigned will still be at a college level.
· Wilco instructors may drop students from Dual Credit if they are not meeting the academic requirements of the class with a grade of ‘C’ or higher or 80%.
· A student who does not perform at a passing rate in a dual credit course may jeopardize their college financial aid.
· Some of the credit offered is transferrable to other institutions. Please visit http://www.itransfer.org/ for
more information on transferring credits from one institution to another.

Course Outline & Competency:
Students will participate in classroom and lab assessments to gauge Technical Understanding and Shop Task proficiency. Lab assessments will be based on SMAW welds performed with 3/32” & 1/8” E6010 & E7018 electrodes. Proficiency will be demonstrated on A36 Mild Carbon Steel, ranging in thickness from 1/4” to 3/8”. WLD-2 students shall develop proficiency in the 1G, 2G and 2F welding positions during the first semester. In Second Semester students will work in the 3G, 4G and 4F positions, as well as Out of Position Welds. Students will develop a working understanding of Welding Procedure Specifications, Welder Qualification and the process of obtaining an AWS Welder Qualification (Certificate).
In order to receive full credit for the course, students are required to meet minimum degrees of proficiency & demonstrate competency in a variety of assigned tasks per JJC guidelines.

WLD-2 includes the following areas of study and associated Competency Tasks:
Semester 1: WLDG-110, Topics & Tasks

●■
SMAW Welding in the Flat Position (Surfacing, Fillet Welds, Groove Welds)
●■
SMAW Welding in the Horizontal Position (Surfacing, Fillet Welds, Groove Welds)
●■
Oxy/Fuel Cutting & Scarfing
●■
Destructive Testing
●■
Weld Discontinuities and Failures
●■
Welding Procedure Specifications (WPS), Welding Essential Variables, Welder Performance Qualification Standards, Weld Inspection Criteria
●■
Mid-term Exam
●■
AWS D1.1 Welding Code & Qualification Procedures
●■
AWS 2G with backing strip (■” 7018)
●■
AWS 2G open root, no backing (■” 6010 & 7018)
●■
Written final exam

Semester 2: WLDG-120, Topics & Tasks:

●■
Selecting Electrodes: Type & Diameter
●■
Discussion on SMAW Welding in the Vertical Position
■■
Practice padding with E6010 and E7018 electrodes (Vertical Position)
■■
Practice lap joint E6010 and E7018 electrodes
●■
Discussion on SMAW Welding in the Overhead Position
■■
Practice tee joint E6010 and E7018 electrodes
●■
Discussion on Destructive Testing
■■
Practice AWS qualification for 3G with backing strip
●■
Discussion on Weld Discontinuities and Failures
■■
Practice AWS qualification for 3G with backing strip
●■
Discussion on Welding Procedure Qualification and review for mid-term exam
■■
Practice & Perform AWS 3G open root.
●■
Mid-term exam
●■
Discussion on fit up for 4G test
■■
Practice & Perform AWS 4G with backing strip
●■
Discussion on ASME section IX code (4G w/o backing strip)
■■
Practice AWS qualification for 4G open root.
●■
Discussion on 3G open root procedure
■■
AWS qualification for 4G open root.
●■
Written final exam